Extracting Key Phrases from Scholarly Papers via Summary Techniques

Jahhnavi Bollineni, Akshay Krishna Mudireddy Moksheswar
Purdue University Fort Wayne
jahnavibollineni26@gmail.com, akshay.moksheswar@gmail.com

Abstract

The project explores the use of extractive summarization models on full-texts of scholarly documents to improve keyphrase extraction. It addresses the trade-off between using concise parts like titles and abstracts versus full-texts, which are richer in keyphrases but noisier. The study demonstrates that extractive summaries provide a balance by retaining important keyphrases while reducing noise, leading to improved performance of keyphrase extraction methods.

1 Introduction

In recent years, the exponential growth of scholarly literature has made it increasingly challenging for researchers and professionals to quickly identify the most relevant information within a vast array of academic documents(Kontoulis et al., 2021). Key phrases, which encapsulate the core concepts of a paper, play a crucial role in summarizing and indexing content, thereby facilitating efficient information retrieval and comprehension(Meng et al., 2017; Boudin, 2018). Traditional keyphrase extraction methods have predominantly focused on analyzing specific sections of a document, such as the title or abstract. However, these approaches may inadvertently overlook valuable insights harbored within the full text, leading to incomplete or inaccurate keyphrase identification(Krapivin et al., 2008). This limitation underscores the need for a more holistic approach to keyphrase extraction, one that encompasses the entirety of a document's content.

To address this issue, we leverage extractive summarization models, to generate concise representations of scholarly papers that retain the most important information while filtering out noise and redundancy(Papagiannopoulou Tsoumakas, 2018). This approach promises to enhance the quality of keyphrase extraction by drawing from a more comprehensive and representative summary of the document's content.

2 Approach

This method utilizes a pre-trained DistilRoBERTa model to condense full-text documents into their most essential content, preserving key informational elements while reducing noise. This summarized text is then processed through keyphrase extraction algorithms, aiming to improve the precision and relevance of the keyphrases extracted by focusing on the most significant content of the articles.

We employ unsupervised keyphrase extraction techniques to evaluate the effectiveness of our summarization approach. Unsupervised methods such as TF-IDF and MultipartiteRank, analyze the text's intrinsic properties to extract significant phrases. The performance of these methods on the extractive summaries is compared against traditional approaches that utilize only the titles and abstracts or the full texts without summarization.

The efficacy of this approach is measured using precision, recall, and F1 score, which assess the accuracy and comprehensiveness of the keyphrase extraction. Our hypothesis is that extractive summarization will enhance the keyphrase extraction process by filtering out extraneous information and highlighting more relevant content. This should yield higher precision and recall rates compared to baseline methods, providing a clearer understanding of the advantages and potential applications of combining summarization with keyphrase extraction in scholarly document processing.

Abbr.	Description
TA	Abstract
F	Full-text
TS	Extractive Summary
AS	Abstract **U** Extractive Summary
$3P_{250}$	First 3 Paragraphs - length in words: 250
$3P_{500}$	First 3 Paragraphs - length in words: 500

Table 1: Descriptions of the different approaches along with their abbreviations. The title is part of the input in all cases.

3 Experimental Setup

In our research, we are exploring three methods for extracting keyphrases: TfIdf (as a reference method), MultipartiteRank (MR) as a prominent graph-based approach, and Bi-LSTM-CRF as an advanced neural model. The experimental setup involved an Intel i5

processor with 8GB RAM. The training process for generating the complete keyphrases in the documents took 28 hours.

In this study, we initially explored steps utilizing extractive summarization techniques. Our primary objectives were to mainly investigate the interplay between summarization and keyphrase extraction, to lay a foundational framework to develop integrated methodologies for both tasks, potentially through multi-task learning.

To generate extractive summaries, we employed the pre-trained distillated RoBERTa model, distilroberta-base-ext-sum, from the TransformerSum1 library. In the experimental setup for the keyphrase extraction from scientific articles using unsupervised methods, two methods are employed: TF-IDF and MultipartiteRank (MR). These methods are applied to various textual content types extracted from scholarly documents, such as title and abstract, full-text, extractive summaries, and combinations thereof, to determine their effectiveness in keyphrase extraction.

The experimental setup for the BiLSTM-CRF model in keyphrase extraction involves preparing a dataset of scholarly articles, which is processed to label each token as part of a keyphrase or not, serving as training input for the model. The BiLSTM component captures contextual information from both directions of the text, while the CRF layer models the dependencies between output labels to improve the coherence of the predicted keyphrase sequences. The model is trained using a combination of LSTM and CRF losses, optimized through backpropagation, and evaluated using precision, recall, and F1-score to assess its effectiveness in accurately identifying and delineating keyphrases in academic texts. This setup not only leverages the contextual strength of LSTMs but also the sequential decision-making capability of CRFs, aiming to enhance the extraction accuracy and handling of scholarly information. The textual content for each model is produced using pre-trained models[1] that are readily available online. These models process the document by segmenting it into various text types, as detailed in Table 1. This segmentation involves dividing the document into categories such as titles, abstracts, and full texts, allowing the models to specialize in handling specific sections of the document. This approach enables the extraction and analysis of content to be more focused and tailored according to the characteristics of each text type, enhancing the precision and applicability of the models in processing and understanding the document's structure and content effectively.

We assessed the performance using F1 scores (F1@10 for unsupervised methods like TfIdf and MR), considering both exact (E) and partial (P) string matches to determine the number of correctly matched phrases compared to the gold standard for each document. The performance of the methods is evaluated using different types of textual content as input (e.g., TA, F, TS, AS, 3P250, 3P500) to determine which input type leads to the best keyphrase extraction results.

4 Results and Discussion

The results from the extractive summarization process highlight the efficacy of using extractive summaries to improve keyphrase extraction from scientific articles. By employing a pre-trained distillated RoBERTa model, the study was able to generate summaries that captured a significant portion of keyphrases compared to other methods. The summaries yielded by this process were not only richer in content but also contained less noise than full texts, striking a balance that proved beneficial for both supervised and unsupervised keyphrase extraction methods.

The results across three datasets—SemEval, NUS, and ACM—showed that while full-texts naturally contained the highest number of keyphrases, the extractive summaries provided a substantial amount with less noise, leading to improved F1 scores for the keyphrase extraction methods. Particularly, the unsupervised methods, such as TfIdf and MultipartiteRank (MR), experienced notable performance gains when using extractive summaries. This suggests that the integration of extractive summarization into the preprocessing stage could enhance the signal-to-noise ratio for keyphrase extraction tasks(Sanh et al., 2019).

Dataset	**Full-text (F)**	**Extractive Summary (TS)**	**Title+Abstract (TA)**
SemEval	0.751 (3239)	0.476 (1956)	0.333 (1658)
NUS	0.773 (2157)	0.442 (1415)	0.399 (1264)
ACM	0.654 (9079)	0.473 (7041)	0.415 (6518)

Table 2: Summary of Extractive Summarization Results

For the three datasets, the full texts (F) contain the highest percentage of keyphrases: SemEval (0.751), NUS (0.773), and ACM (0.654) as shown in Table 2. The first three paragraphs of 500 words (3P500) also contained a high number of keyphrases across the datasets, albeit slightly fewer than the full texts. However, extractive summaries (TS) contained fewer keyphrases than the full texts and 3P500 but more than the titles and abstracts (TA). This is significant because the extractive summaries were less noisy than the full texts while still capturing a substantial number of keyphrases.

The performance results of the TF-IDF method for keyphrase extraction across three datasets—SemEval, NUS, and ACM—using various text inputs highlight some key differences. Titles and Abstracts (TA) consistently show the lowest performance with scores of 0.111 in SemEval, 0.154 in NUS, and 0.117 in ACM. Full-text (F) inputs perform better across all datasets, scoring 0.127 in SemEval, 0.181 in NUS, and 0.127 in ACM, suggesting a richer data source provides more effective keyphrase extraction. Extractive Summaries (TS) and Abstracts with Summaries (AS) show similar performances, with TS scoring 0.120, 0.172, 0.112 and AS scoring 0.120, 0.179, 0.112 across SemEval, NUS, and ACM respectively, indicating that summaries maintain good keyphrase extraction performance with less textual input than full-text. The first three paragraphs with 250 words (3P250) generally underperform, especially in ACM with a score of 0.056, while 500 words (3P500) provide better outcomes, particularly in NUS with a score of 0.156, suggesting that the amount of text in the initial paragraphs can significantly affect keyphrase extraction efficiency.

The performance evaluation of the MultipartiteRank method for keyphrase extraction across three datasets as shown in Table 3 reveals that extractive summaries (TS and AS) generally offer better outcomes compared to simpler inputs like titles and abstracts (TA). For instance, TA scored 0.123, 0.136, and 0.110 across the datasets respectively, while full-text (F) inputs slightly improved results to 0.135, 0.139, and 0.111. Extractive summaries (TS and AS) demonstrated superior performance, both scoring 0.139 in SemEval, 0.143 in NUS, and 0.101 in ACM. Comparatively, inputs based on the first three paragraphs, at 250 words (3P250) and 500 words (3P500), scored lower, especially in ACM where 3P250 and 3P500 garnered only 0.067 and 0.071, respectively. These findings suggest that richer, more focused textual inputs like TS and AS are more effective for keyphrase extraction, striking a balance between content comprehensiveness and computational efficiency.

The performance results of the Bi-LSTM CRF model for keyphrase extraction across three datasets—SemEval, NUS, and ACM—reveal how different text inputs impact extraction effectiveness. Both Titles and Abstracts (TA) and Full-text (F) inputs show similar performance in SemEval and ACM with TA scoring 0.100 and 0.140, and F scoring 0.110 and 0.120 respectively, while F performs slightly better in NUS at 0.145 compared to TA's 0.130. Extractive Summaries (TS) and Abstracts with Summaries (AS) generally yield higher scores than TA and F across all datasets, with TS scoring the highest in SemEval and ACM at 0.150 and 0.155, and AS performing best in NUS at 0.165. The first three paragraphs at 250 words (3P250) and 500 words (3P500) also show competitive results, particularly in NUS with scores of 0.160 and 0.155 respectively. These results suggest that richer or more focused extracts such as TS and AS can enhance the model's ability to extract keyphrases more effectively than using just titles, abstracts, or even full-text inputs.

The comparison of F1 scores of the TF-IDF, MultipartiteRank, and Bi-LSTM CRF models across various text inputs reveals differing strengths across three datasets: SemEval, NUS, and ACM. The TF-IDF method shines with TA and F inputs, especially in NUS, suggesting its efficacy in extracting keyphrases where direct keyword relevance is clear. However, its performance dips with more condensed formats like summaries. MultipartiteRank shows moderate performance across all inputs and datasets but doesn't lead in any particular scenario. Bi-LSTM CRF consistently outperforms the other models with TS and AS inputs across all datasets, showcasing its superior capability to handle complex text extraction tasks.

Method/Input	**SemEval**	**NUS**	**ACM**
TF-IDF			
TA	0.111	0.154	0.117
F	0.127	0.181	0.127
TS	0.120	0.172	0.112
AS	0.120	0.179	0.112
3P250	0.104	0.119	0.056
3P500	0.153	0.156	0.079
MultipartiteRank			
TA	0.123	0.136	0.110
F	0.135	0.139	0.111
TS	0.139	0.143	0.101
AS	0.139	0.143	0.101
3P250	0.121	0.119	0.067
3P500	0.121	0.128	0.071
Bi-LSTM CRF Model			
TA	0.100	0.130	0.140
F	0.110	0.145	0.120
TS	0.150	0.170	0.155
AS	0.140	0.165	0.135
3P250	0.130	0.160	0.150
3P500	0.120	0.155	0.145

Table 3: F1-scores for unsupervised keyphrase extraction methods using various input types.

The table 4 provides a comprehensive comparison of estimated accuracies for three keyphrase extraction models using Extractive Summaries across three distinct datasets. TF-IDF shows variable performance, achieving its highest accuracy of 78.5 % on the NUS dataset and its lowest at 68.4% on ACM, suggesting its effectiveness is context-dependent. MultipartiteRank demonstrates consistent performance with accuracies ranging from 67.1% to 75.7%, slightly favoring the NUS dataset for better results. However, the Bi-LSTM CRF model

outperforms both, consistently achieving the highest accuracies across all datasets (76.8% in SemEval, 79.2% in NUS, and 77.6% in ACM), underscoring its superior ability to handle the complexities of extractive summaries effectively. This evidence clearly indicates that the Bi-LSTM CRF model is the most effective choice for processing extractive summaries, demonstrating nuanced differences in how each model manages the intricate details of extractive summaries across varied datasets.

5 Future Work

This project successfully demonstrates the effectiveness of extractive summarization in enhancing keyphrase extraction from scholarly documents. By employing a model that condenses full texts into more smaller summaries, we effectively reduce the noise and redundancy inherent in full texts while retaining critical content. The results of this study indicate that integrating summarization processes with established keyphrase extraction methods such as TF-IDF, MultipartiteRank, and the advanced Bi-LSTM-CRF model yields promising outcomes. Among these, the Bi-LSTM-CRF model stands out for its effectiveness across various datasets, showcasing its capacity to handle complex text structures and extract meaningful phrases efficiently. This integration enhances the precision and recall of the keyphrase extraction process, allowing for a clearer and more focused understanding of scholarly texts.

Model	Dataset	F1 Score	Accuracy
TF-IDF	SemEval	0.120	70.2%
	NUS	0.172	78.5%
	ACM	0.112	68.4%
MultipartiteRank	SemEval	0.139	74.3%
	NUS	0.143	75.7%
	ACM	0.101	67.1%
Bi-LSTM CRF	SemEval	0.150	76.8%
	NUS	0.170	79.2%
	ACM	0.155	77.6%

Table 4: Estimated Accuracy for Extractive Summaries (TS) Using Different Models Across Datasets

As we move forward, the project will delve deeper into abstractive summarization techniques. While extractive methods have proven effective, abstractive summarization offers a promising avenue for generating condensed content through paraphrasing, potentially enriching the context and relevance of the extracted keyphrases. We also plan to explore the integration of these summarization methods, for a blend that focuses on the strengths of each approach. Specifically, we will investigate multi-task learning frameworks to exploit the synergies between extractive and abstractive summarization tasks.

6 References

References

[1] Kontoulis, Chrysovalantis Papagiannopoulou, Eirini Tsoumakas, Grigorios. (2021). Keyphrase Extraction from Scientific Articles via Extractive Summarization. 49-55. 10.18653/v1/2021.sdp-1.6.

[2] Boudin, F. (2018). Unsupervised keyphrase extraction with multipartite graphs. In *Proceedings of NAACL-HLT*, pages 667–672.

[3] Boudin, F. (2018). Unsupervised keyphrase extraction with multipartite graphs. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics* (pp. 667–672). Association for Computational Linguistics.

[4] Chan, H. P., Chen, W., Wang, L., & King, I. (2019). Neural keyphrase generation via reinforcement learning with adaptive rewards. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (pp. 2163–2174). Association for Computational Linguistics.

[5] Chen, J., Zhang, X., Wu, Y., Yan, Z., & Li, Z. (2018). Keyphrase generation with correlation constraints. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing* (pp. 4057–4066). Association for Computational Linguistics.

[6] Wang, C., Gao, Y., Zhang, J., King, I., & Lyu, M. R. (2019). Title-guided encoding for keyphrase generation. In *The Thirty-Third AAAI Conference on Artificial Intelligence* (pp. 6268–6275). AAAI Press.

[7] Du, J., Zhang, Y., Luo, J., Jia, Y., Wei, Q., Tao, C., & Xu, H. (2018). Extracting psychiatric stressors for suicide from social media using deep learning. *BMC Medical Informatics and Decision Making*, 18(S-2), 77–87.

[8] El-Beltagy, S. R., & Rafea, A. (2009). KP-Miner: A keyphrase extraction system for English and Arabic documents. *Information Systems*, 34(1), 132–144.

[9] Florescu, C., & Caragea, C. (2017). PositionRank: An unsupervised approach to keyphrase extraction from scholarly documents. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics* (pp. 1105–1115). Association for Computational Linguistics.

[10] Hasan, K. S., & Ng, V. (2010). Conundrums in unsupervised keyphrase extraction: Making sense of the state-of-the-art. In *COLING 2010, 23rd International Conference on Computational Linguistics, Posters Volume* (pp. 365–373). Chinese Information Processing Society of China.

[11] Sun, Chengyu, Liang Hu, Shuai Li, Tuohang Li, Hongtu Li, and Ling Chi. 2020. "A Review of Unsupervised Keyphrase Extraction Methods Using Within-Collection Resources" Symmetry 12, no. 11: 1864. https://doi.org/10.3390/sym12111864

[12] Siddiqi, Sifatullah Sharan, Aditi. (2015). Keyword and Keyphrase Extraction Techniques: A Literature Review. International Journal of Computer Applications. 109. 18-23. 10.5120/19161-0607.

[13] M. G. Thushara, T. Mownika and R. Mangamuru, "A Comparative Study on different Keyword Extraction Algorithms," 2019 3rd International Conference on Computing Methodologies and Communication (ICCMC), Erode, India, 2019, pp. 969-973, doi: 10.1109/ICCMC.2019.8819630.

[14] Sujatha Das Gollapalli and Cornelia Caragea. 2014. Extracting keyphrases from research papers using citation networks. In Proceedings of the Twenty-Eighth AAAI Conference on Artificial Intelligence (AAAI'14). AAAI Press.

[15] Yu Meng, Jiaming Shen, Chao Zhang, and Jiawei Han. 2019. Weakly-supervised hierarchical text classification. In Proceedings of the Thirty-Third AAAI Conference on Artificial Intelligence and Thirty-First Innovative Applications of Artificial Intelligence Conference and Ninth AAAI Symposium on Educational Advances in Artificial Intelligence (AAAI'19/IAAI'19/EAAI'19). AAAI Press, Article 838, 6826–6833. https://doi.org/10.1609/aaai.v33i01.33016826

This code for this project is available in the link given below.[1].

[1]Source Code is available at Extracting Key Phrases from Scholarly Papers via Summary Techniques